

A DGS-based Guitar-shaped Antenna for WiFi-2.4 GHz/ ZigBee/Bluetooth/ISM Band Applications

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Abstract— This paper presents a defected ground structure (DGS)-based guitar-shaped microstrip patch antenna (GSPA) for 2.4 GHz applications. The GSPA is designed on a compact Rogers RT5880 dielectric substrate with dimensions of $32 \times 38 \times 1.575$ mm³. To enhance bandwidth and impedance matching, a slotted parasitic element and a slotted ground structure are incorporated into the design. It achieves a bandwidth of 730 MHz (2.25-2.98 GHz) with a return loss of -38.48 dB and a maximum efficiency of approximately 97%. The compact footprint, simple structure, stable gain, and symmetrical radiation pattern make the GSPA a promising candidate for 2.4 GHz WiFi, Zigbee, Bluetooth, and ISM band. A comparative analysis with existing works further demonstrates that the design offers competitive performance in terms of size, efficiency, bandwidth, and overall radiation characteristics.

Keywords— GSPA; DGS; WiFi; ZigBee; Bluetooth; ISM Band.

I. INTRODUCTION

The rapid integration of advanced technologies in modern wireless systems has significantly increased the demand for compact, flexible, and high-performance antenna systems. Over the past few decades, the number of devices and systems operating in the Industrial, Scientific, and Medical (ISM) frequency bands has grown remarkably [1]. The Federal Communications Commission (FCC) has allocated several frequency bands for ISM applications, including 433.05–434.79 MHz, 902–928 MHz, 2.4–2.5 GHz, and 5.725–5.875 GHz [2]. In addition to their extensive use in industrial and scientific applications, ISM bands are widely employed in medical devices and wireless communication technologies such as International Mobile Telecommunications (IMT) at 2.3–2.4 GHz, Zigbee at 2.4 GHz, Bluetooth at 2.402–2.48 GHz, Wi-Fi at 2.412–2.4835 GHz, WiMAX at 2.30–2.40 GHz and 2.496–2.69 GHz, WLAN at 2.4–2.483 GHz, and wireless sensor networks [2], [3].

Although lower ISM frequency bands offer longer communication ranges and are suitable for various applications, they generally require antennas with larger physical dimensions. This makes them less practical for implantable medical devices, compact wireless modules, and portable handheld systems. Therefore, the 2.45 GHz ISM band has become a preferred operating range because it enables antenna miniaturization while maintaining reliable and stable performance [4]. Consequently, the development of compact planar antennas for narrowband ISM applications has become essential for current and future wireless communication systems. Such antennas must be lightweight, compact, and easy to integrate with portable and embedded platforms. In addition, they should offer high radiation efficiency, stable impedance matching, and consistent radiation performance to satisfy the requirements of high-speed, low-latency, and reliable communication services.

Several microstrip patch antennas operating in the 2.4 GHz ISM-band have been reported in the literature for WLAN, Wi-Fi, and Internet of Things (IoT) applications. A compact PSI-shaped patch antenna with dimensions of $38.04 \times 38.04 \times 0.79$ mm³ was proposed for WLAN applications in [5], achieving a radiation efficiency of approximately 70% using an Arlon AD270N substrate. Conventional microstrip patch antennas for Wi-Fi applications were presented in [6], and [7], both resonating around 2.4 GHz, confirming the suitability of traditional patch geometries for ISM-band operation. Studies in [7] and [8] further focused on improving impedance matching and radiation characteristics. In particular, K. P. Lad et al. reported a compact antenna with a return loss of -52.89 dB at 2.4 GHz [7], while an IoT-oriented antenna in [8] achieved a limited impedance bandwidth of 60 MHz. To extend antenna functionality, dual-frequency microstrip patch antennas were proposed in [9], operating at two distinct frequency bands on FR-4 substrates with high return loss. However, these dual-band designs required relatively large antenna footprints, reaching up to $85 \times 85 \times$

1.6 mm³ in [9], which limits their suitability for compact and wearable IoT devices. Furthermore, several FR-4-based designs reported radiation efficiencies below 75%, as observed in [6], [9], [10], and [11], demonstrating the common trade-off between low-cost substrates and radiation performance. In contrast, antennas fabricated on low-loss substrates, such as the Arlon-based design in [5], can provide improved efficiency. Additionally, the circular patch antenna presented in [12] achieved a maximum gain of 4.40 dB using a 70 × 70 × 3.6 mm³ FR-4 substrate, indicating its potential for short-range IoT applications.

In this study, a DGS-based GSPA is proposed for 2.4 GHz ISM-band. The GSPA is designed on a compact Rogers RT5880 substrate, and annealed copper is used as the conductive material. The GSPA provides satisfactory gain, directivity, and radiation efficiency over a wide impedance bandwidth of 730 MHz, covering the frequency range of 2.25–2.98 GHz. To improve impedance matching and enhance overall performance, slots are introduced into the radiating patch, parasitic elements, and partial ground plane. The use of a slotted parasitic together with a DGS contributes to bandwidth enhancement and stable radiation characteristics. The rest sections of this paper is organized as follows. Section II describes the antenna design and structural configuration. Section III presents the results and analysis. Finally, Section IV concludes the paper.

II. DESIGN AND STRUCTURE OF GSPA

The GSPA design and analysis of the parametric result are presented in this part. Time Domain (TD) analysis using CST Microwave Studio is used to build and simulate the GSPA based on DGS. Optimization is performed using a low loss dielectric substrate Rogers RT5880 having thickness of 1.575 mm with a slotted parasitic ground plane. As shown in Fig. 1, the suggested guitar-shape antenna is made up of a patch with a circular and two perpendicular line slots, two parasitic squares, and a 50 Ω feedline. Initially, the dimension of antenna was calculated using a few basic formulae (1), (2), (3), and (4), and the antenna was optimized to 32×38×1.575 mm³ [13]. Geometrical parameter details are listed in Table I.

Patch width,

$$W_p = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}} \quad (1)$$

Length,

$$L = 0.412 \frac{(\epsilon_{reff} + 0.3)(\frac{W_p}{h} + 0.264)}{(\epsilon_{reff} - 0.258)(\frac{W_p}{h} + 0.8)} \quad (2)$$

Effective length,

$$L_{eff} = \frac{c}{2f_r \sqrt{\epsilon_{reff}}} \quad (3)$$

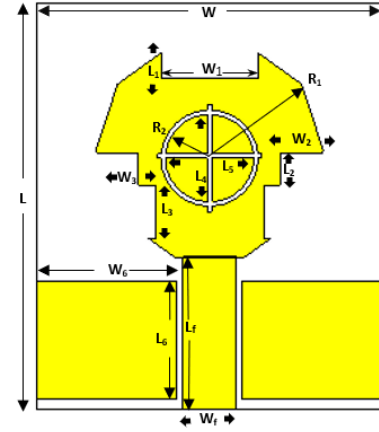
Patch length,

$$L_p = L_{eff} - 2\Delta L \quad (4)$$

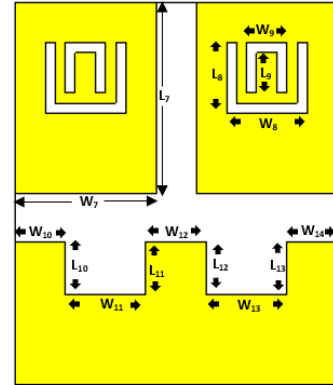
where, c = velocity of light, f_r = resonance frequency, ϵ_r = relative permittivity and h = height of the substrate.

The length and width of the feed line are $L_f = 14.30$ mm and $W_f = 4.94$ mm, respectively. The proposed design incorporates two types of parasitic elements, which are positioned on the dielectric substrate and the ground plane. The parasitic elements placed on the dielectric substrate have a length of $L_6 = 11$ mm and a width of $W_6 = 13$ mm. In contrast, the parasitic elements introduced on the ground plane

have a length of $L_7 = 19$ mm and a width of $W_7 = 14$ mm, as illustrated in Fig. 1. A plus shaped and circular slots are etched on the radiating patch. The difference between the outer and inner radii of the circular slot is 0.40 mm, and the same dimensional spacing is maintained for the plus-shaped slot. Two parasitic elements are also introduced on the ground plane, each containing two slots with the same configuration. For the slots on the ground parasitic elements, the gap between the inner and outer slots is 1 mm. The parasitic elements and slots are incorporated to improve impedance matching, gain, and directivity of the DGS-based patch antenna. These structural modifications help enhance the overall antenna performance, making the design a suitable candidate for 2.4 GHz Wi-Fi, Zigbee, Bluetooth, and ISM-band.



(a) Front view



(b) Back view

Fig. 1. DGS-based GSPA.

TABLE I. DIMENSION DETAILS OF THE GSPA

Description	Value (mm)	Description	Value (mm)
Length, L	38	Ground parasite width, W ₇	14.00
Width, W	32	Ground parasitic slot length, L ₈	7.00
Patch radius, R ₁	10.55	Ground parasitic slot width, W ₈	8.00
Slot radius, R ₂	4.10	Ground Parasitic slot, L ₉ = W ₉	4.00
Patch slot length-1, L ₁	2.32	Ground plane slot width-1, W ₁₀	5.00
Patch slot width-1, W ₁	9.00	Ground plane slot length-1, L ₁₀	5.25
Patch slot length-2, L ₂	3.00	Ground plane slot width-2, W ₁₁	8.00
Patch slot width-2, W ₂	3.92	Ground plane slot length-2, L ₁₁	5.25
Patch slot length-3, L ₃	5.00	Ground plane slot width-3, W ₁₂	6.00
Patch slot width-3, W ₃	1.60	Ground plane slot length-3, L ₁₂	5.25
Patch slot length-4, L ₄	10.00	Ground plane slot width-4, W ₁₃	8.00
Patch slot length-5, L ₅	9.60	Ground plane slot length-4, L ₁₃	5.25
Ground parasitic length, L ₇	19.00	Ground plane slot width-4, W ₁₄	5.00

III. RESULTS AND ANALYSIS

The performance of the compact GSPA is evaluated in terms of reflection coefficient ($|S_{11}|$), VSWR, impedance matching, directivity, gain, radiation efficiency, field distribution, and surface current concentration. The simulated scattering parameter ($|S_{11}|$) of the GSPA is presented in Fig. 2. It exhibits a good resonance at 2.54 GHz with a $|S_{11}|$ of -38.48 dB. Based on the -10 dB impedance bandwidth criterion, the antenna achieves a bandwidth of 730 MHz, covering 2.25 to 2.98 GHz. The low reflection coefficient value of -38.48 dB indicates that only a very small portion of the incident power is reflected back toward the transmission line, while most of the power is effectively delivered to the antenna. This confirms good impedance matching between the GSPA and the transmission line and contributes to a low VSWR. As shown in Fig. 3, the VSWR remains between 1 and 2 across the entire operating bandwidth, indicating acceptable impedance matching throughout the band. At the resonant frequency, the VSWR is 1.0243, which demonstrates that the impedance is very closely matched to the standard $50\ \Omega$ transmission line.

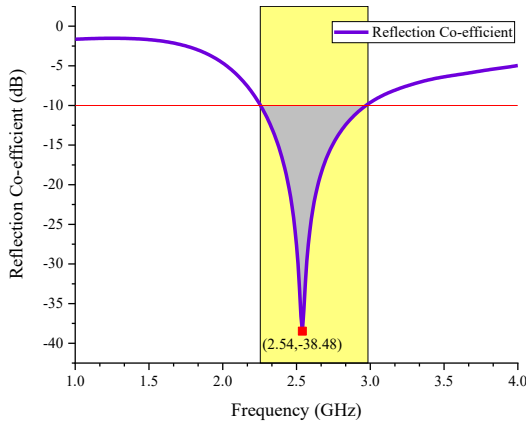


Fig. 2. $|S_{11}|$ curve of the designed compact GSPA.

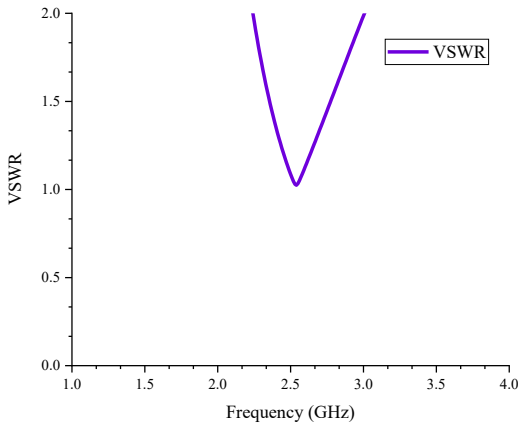


Fig. 3. VSWR analysis of the compact GSPA.

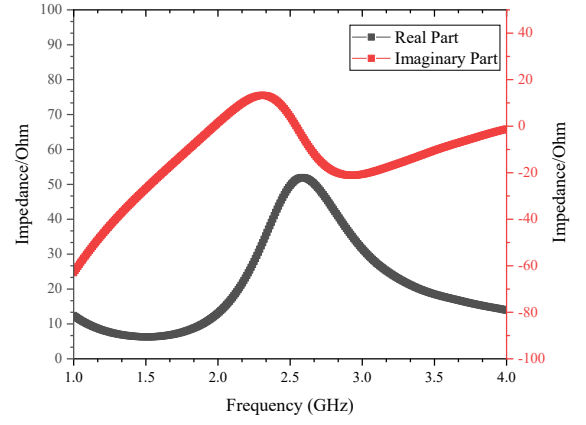


Fig. 4. Impedance analysis of the compact GSPA.

The impedance characteristics of the GSPA are illustrated in Fig. 4. At 2.54 GHz, the real part of the input impedance is close to $50\ \Omega$, with a value of $51.299\ \Omega$, while the imaginary part is nearly $0\ \Omega$, with a value of $0.18935\ \Omega$. This indicates that the GSPA impedance is well matched to the transmission line, allowing maximum radio frequency (RF) power to be transferred from the transmission line to the antenna with minimal reflection toward the source. The radiation performance of the proposed antenna is presented in Fig. 5. The antenna maintains a high radiation efficiency of more than 93% across the entire operating range of 2.25–2.98 GHz. At 2.54 GHz, the antenna achieves a peak radiation efficiency of 97%, demonstrating its suitability for practical systems in the ISM band and related communication systems. The frequency-dependent comparison of directivity and gain over the operating band is shown in Fig. 6. The antenna achieves a maximum directivity of 2.73 dBi and a maximum gain of 2.58 dB within the operating band. At 2.54 GHz, the antenna exhibits a directivity of 2.602 dBi and a gain of 2.298 dB. The corresponding 3D gain and directivity plots of the proposed GSPA are illustrated in Fig. 7(a-b).

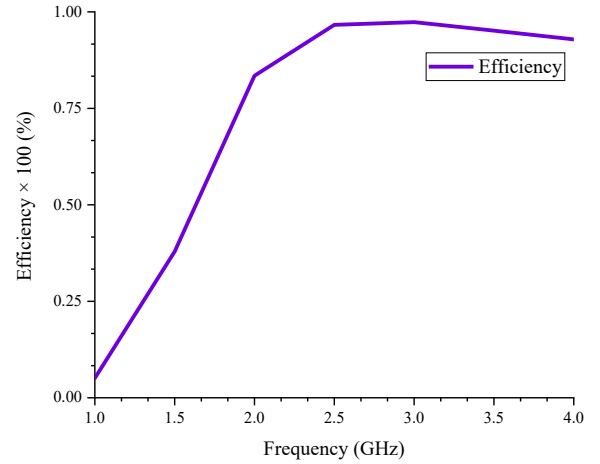


Fig. 5. Radiation efficiency of the GSPA.

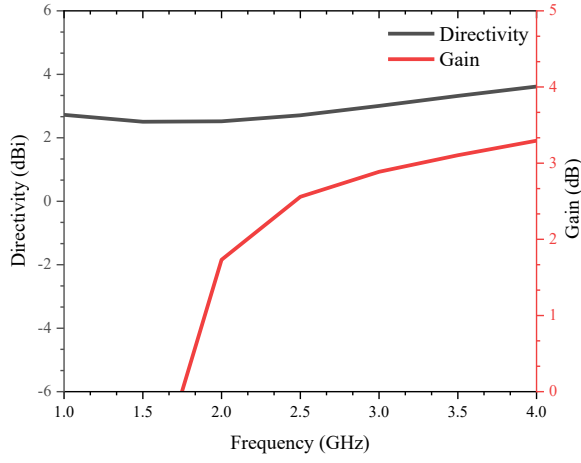


Fig. 6. Gain and directivity curve of the designed compact GSPA.

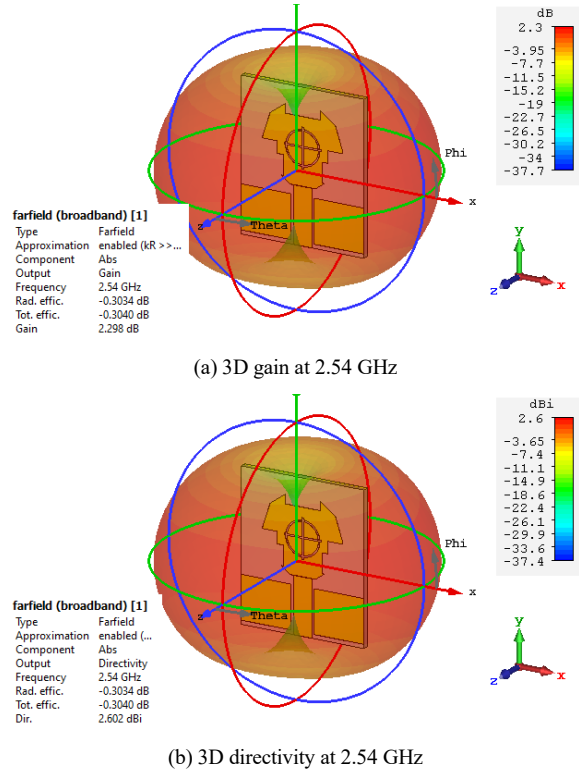


Fig. 7. 3D radiation pattern of the designed compact GSPA.

The polar radiation patterns of the GSPA, including both E- and H-field distributions at $\phi = 0^\circ$ and $\phi = 90^\circ$, are presented in Fig. 8 at 2.54 GHz. For the E-field pattern, shown in Fig. 8(a), the main lobe is directed approximately toward $\phi = 90^\circ$, with a magnitude of 17.1 dBV/m for both $\phi = 0^\circ$ and $\phi = 90^\circ$. Similarly, the H-field pattern in Fig. 8(b) shows that the major lobe is also radiated around $\phi = 180^\circ$, with a main lobe magnitude of -34.5 dBA/m for $\phi = 0^\circ$ and $\phi = 90^\circ$. In both field patterns at $\phi = 90^\circ$, a side lobe level of approximately -1.0 dB is observed, with an angular width of 88.4° . The surface current distribution indicates strong current dispersion across the lower portion of the radiating patch. At 2.54 GHz, the maximum surface current density reaches 54.8416 A/m, as shown in Fig. 9. The overall performance of the GSPA is summarized in Table II. Furthermore, Table III compares the

proposed GSPA with other published works, demonstrating that the design offers a compact volume and improved bandwidth. In addition, the antenna achieves a maximum radiation efficiency of approximately 97%, confirming its suitability for 2.4 GHz ISM-band systems.

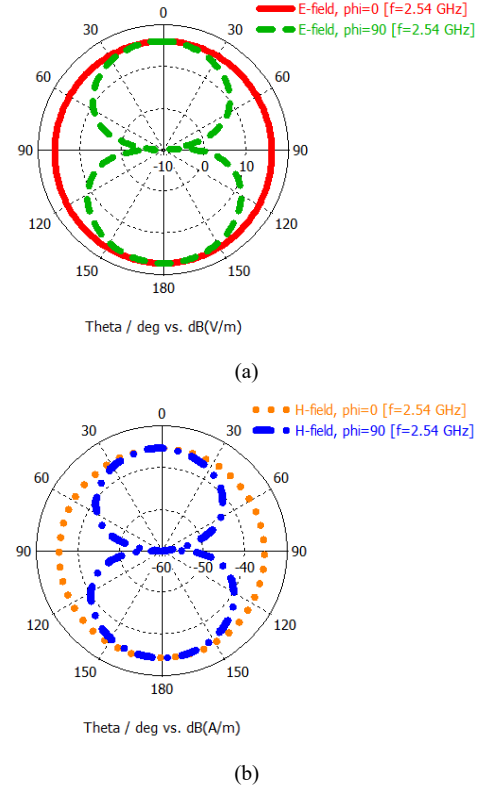


Fig. 8. (a) E-field and (b) H-field analysis of the GSPA.

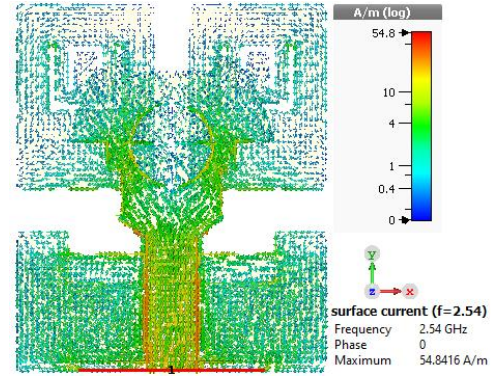


Fig. 9. Current distribution at 2.54 GHz.

TABLE II. PERFORMANCE METRICS OF THE GSPA

Parameter Name	Value
Range (GHz)	2.25-2.98
BW (MHz)	730
Resonant Frequency (GHz)	2.54
Return loss (dB) at 2.54 GHz	-38.48
Max. Directivity (dBi)	2.73
Max. Gain (dB)	2.58
Max. Efficiency (%)	97
VSWR	1.0243
Impedance (Ω)	51.299 -j0.18935

TABLE III. COMPARISON WITH RECENT RELAVENT WORKS

Index	Reference No.								This work
	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	
Year	2023	2023	2022	2023	2022	2024	2023	2023	2025
Size (L×W×h) mm ³	38.04×38×0.79	38.48×47.04×1.5	25×20×1.5	47.04×38.48×1.5	85×85×1.6	58×76.74×1.6	-	70×70×3.6	32×38×1.575
Substrate material	Arlon Ad270N	FR-4	FR-4	FR-4	FR-4	FR-4	FR-4	FR-4	Rogers RT5880
Operating Frequency (GHz)	≈2.39–2.41	1–3.5	2.33–2.48	~2.36–2.42	1.215–1.239, 2.4–2.48	2.366–2.4336	2.3346–2.2625, 3.4219–3.372	~2.37–2.5	2.25–2.98
Centre frequency (GHz)	2.4	2.4	2.4	2.4	1.227, 2.4	2.4	2.4, 3.5	2.45	2.54
Return loss (dB)	-32.53	-11.3048	-52.89	-23.942	-18.5	-21.16	-14.24, -12.01	-19.7332	-38.48
BW (MHz)	20	2500	150	60	30, 50	67.6	72, 49	130	730
Maximum Gain (dB)	5.47	1.8787	-	4.01	4.22, 2.39	2.3	1.508, 7.01	4.40	2.54
VSWR	1.048	1.8060	-	1.16	1.8, 2	1.19	1.47, 1.6	1.30	1.0243
Efficiency (%)	70.39	-	86.975	90	-	-	-	92	97
Application	RF harvesting, RFID, WLAN	Wi-Fi	Wi-Fi, WLAN	IoT	GPS, Wi-Fi	Wireless	Wireless	IoT	WiFi-2.4 GHz, ZigBee, Bluetooth, ISM Band

IV. CONCLUSION

This article presents a slotted GSPA with a DGS for Wi-Fi 2.4 GHz, Zigbee, Bluetooth, and ISM-band. The incorporation of rectangular and circular slots on the radiating patch, along with the DGS on the ground plane, effectively enhances the radiation performance and impedance matching of the GSPA. The designed GSPA achieves a strong reflection coefficient of -38.48 dB at 2.54 GHz and provides a wide impedance bandwidth of 730 MHz. Moreover, the VSWR remains within the acceptable range of 1 to 2 over the entire operating bandwidth, confirming good impedance matching. Despite its simple configuration and compact footprint of 32×38 mm², the GSPA exhibits satisfactory gain, directivity, and high efficiency, along with stable omnidirectional radiation characteristics suitable for communication systems. A detailed parametric investigation was also carried out to evaluate the influence of key design parameters on GSPA performance. The antenna performance was verified through CST electromagnetic simulation using the TD and frequency domain (FD) solver, and the simulated results confirmed the reliability and effectiveness of the proposed design. Furthermore, a comparative analysis with previously reported antennas highlights the advantages of the GSPA in terms of compact size, improved bandwidth, high efficiency, and practical applicability. Therefore, the proposed slotted GSPA can be considered a promising candidate for Wi-Fi 2.4 GHz, Zigbee, Bluetooth, and ISM-band applications.

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